

SIMATS ENGINEERING

ASSESSMENT TOOL 1 - High (INDUSTRY PROBLEM SOLVING TASKS)

COURSE CODE: CSA09

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO4: *Design and develop GUI-based user interfaces using Abstract Window Toolkit, Swing, and Applets by implementing event handling, layout managers, AWT controls, and menus. (BL3,BL5)*

Assessment Tool Weightage: 10%

QUESTIONS: 5

Total Marks: 50

ANNEXURE A

INDUSTRY PROBLEM SOLVING TASKS

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

ANNEXURE

INDUSTRY PROBLEM SOLVING TASKS

5. Online Examination System Interface (BTL: BL5 – Evaluate)

Design a Swing-based GUI application for conducting online exams, including features like question navigation, timer display, and result submission. Use components such as radio buttons, panels, and menus. Implement event handling for answering questions and submitting responses. Validate user actions to prevent multiple submissions. Evaluate layout managers for organizing question panels and controls. Analyze usability and responsiveness under multiple users and propose enhancements for reliability and scalability.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts

Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

JAVA CODE:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class OnlineExamApplet extends JApplet implements ActionListener {

    JLabel questionLabel, timerLabel, resultLabel;
    JRadioButton option1, option2, option3, option4;
    ButtonGroup options;
    JButton nextButton, submitButton;

    int question = 1
    int score = 0;
    int time = 60;
    boolean submitted = false;
    Timer timer;

    public void init() {

        setLayout(new BorderLayout())

        JPanel topPanel = new JPanel(new BorderLayout());

        JLabel title = new JLabel(

            "ONLINE EXAMINATION SYSTEM",

            JLabel.CENTER

        );

        title.setFont(new Font("Arial", Font.BOLD, 20));
        timerLabel = new JLabel("Time: 60 seconds");
        topPanel.add(title, BorderLayout.CENTER);
        topPanel.add(timerLabel, BorderLayout.EAST);
        add(topPanel, BorderLayout.NORTH);

        JPanel questionPanel = new JPanel();
        questionPanel.setLayout(new GridLayout(5, 1));
        questionLabel = new JLabel(
```

```
);  
option1 = new JRadioButton("Python");  
option2 = new JRadioButton("Java");  
option3 = new JRadioButton("HTML");  
option4 = new JRadioButton("CSS");  
options = new ButtonGroup();  
options.add(option1);  
options.add(option2);  
options.add(option3);  
options.add(option4);  
questionPanel.add(questionLabel);  
questionPanel.add(option1);  
questionPanel.add(option2);  
questionPanel.add(option3);  
questionPanel.add(option4);  
add(questionPanel, BorderLayout.CENTER)  
  
JPanel bottomPanel = new JPanel();  
nextButton = new JButton("Next");  
submitButton = new JButton("Submit");  
resultLabel = new JLabel("");  
nextButton.addActionListener(this);  
submitButton.addActionListener(this);  
bottomPanel.add(nextButton);  
bottomPanel.add(submitButton);  
bottomPanel.add(resultLabel);  
add(bottomPanel, BorderLayou  
  
JMenuBar menuBar = new JMenuBar();  
JMenu examMenu = new JMenu("Exam");  
JMenuItem startItem = new JMenuItem("Start Exam");
```

```

JMenuItem exitItem = new JMenuItem("Exit");
startItem.addActionListener(this);
exitItem.addActionListener(this);
examMenu.add(startItem);
examMenu.add(exitItem);
menuBar.add(examMenu);
setJMenuBar(menuBar);

timer = new Timer(1000, new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        time--;
        timerLabel.setText("Time: " + time + " seconds");
        if (time <= 0) {
            timer.stop();
            submitExam();
        }
    }
});

timer.start();
}

public void actionPerformed(ActionEvent e) {
    if (e.getSource() == nextButton) {
        if (!submitted)
            checkAnswer();
        question++;
        if (question == 2) {
            questionLabel.setText(
                "Q2. Which keyword is used for inheritance?"
            );
        }
    }
}

```

```

        option1.setText("this");
        option2.setText("extends");
        option3.setText("static");
        option4.setText("final");

    }

    else if (question == 3) {

        questionLabel.setText(
            "Q3. Which Swing component is used to create a button?"
        );
        option1.setText("JLabel");
        option2.setText("JPanel");
        option3.setText("JButton");
        option4.setText("JTextField");

    }

    else {

        JOptionPane.showMessageDialog(
            this,
            "All questions completed. Click Submit."
        );
        nextButton.setEnabled(false);
    }

    options.clearSelection();
}

}

if (e.getSource() == submitButton) {
    submitExam();
}

```

```

    }

    if (e.getActionCommand().equals("Start Exam")) {
        time = 60;
        timer.start();
    }

    if (e.getActionCommand().equals("Exit")) {
        JOptionPane.showMessageDialog(
            this,
            "Exam closed."
        );
    }
}

void checkAnswer() {
    if (question == 1 && option2.isSelected()) {
        score++;
    }

    if (question == 2 && option2.isSelected()) {
        score++;
    }

    if (question == 3 && option3.isSelected()) {
        score++;
    }
}

void submitExam() {
    if (submitted) {
        JOptionPane.showMessageDialog(
            this,
            "Exam already submitted!"
        );
    }
}

```



```
        return;
    }
    checkAnswer();
    submitted = true;
    timer.stop();
    nextButton.setEnabled(false);
    submitButton.setEnabled(false);
    resultLabel.setText(
        "Score: " + score + " / 3"
    );
    JOptionPane.showMessageDialog(
        this,
        "Exam Submitted!\nYour Score: "
        + score + " / 3"
    );
}
}
```

OUTPUT SCREEN SHORT:



